

Physics

1. Convex
2. convex lens of focal length 10cm
3. $u = -5 \text{ cm}$
 $v = 10 \text{ cm}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

convex lens

$$\frac{1}{f} = \frac{1}{10} + \frac{1}{5}$$

$$\frac{1}{f} = \frac{1+2}{10}$$

$$\frac{1}{f} = \frac{3}{10}$$

$$f = \frac{10}{3}$$

$$f = 3.33 \text{ cm}$$

4. $u = -50 \text{ cm}$

Since the size of img is equal to size of obj

The object is at Centre of curvature

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$f = 25 \text{ cm}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{0.25}$$

$$P = 4 \text{ D}$$

